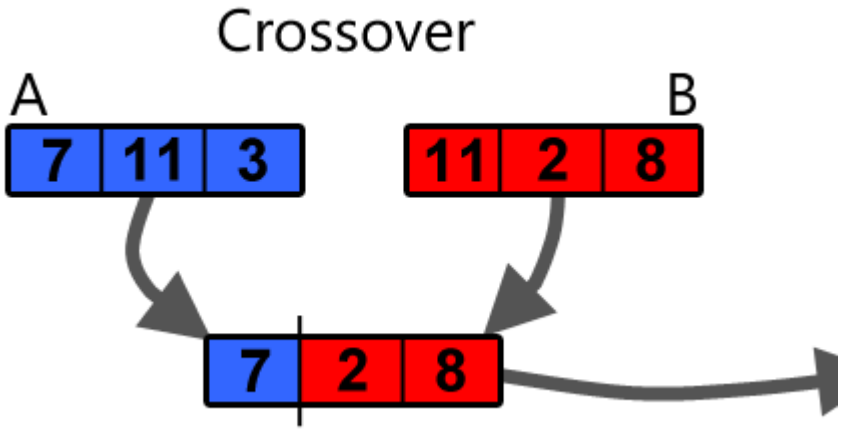
Genetic Algorithms



Exercise 1.

(in groups of 2-3)

Do you agree:

Intelligence = Search + Prediction

(Is Search and Prediction all we need to have in order to be intelligent?)

Genetic Algorithms.

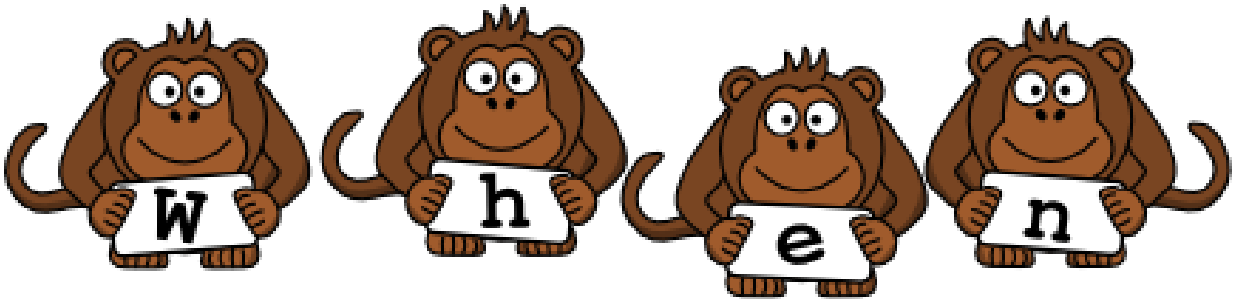
Still:

(Often) You need to find the answer in a pool of possibilities.

(Needle in a haystack)

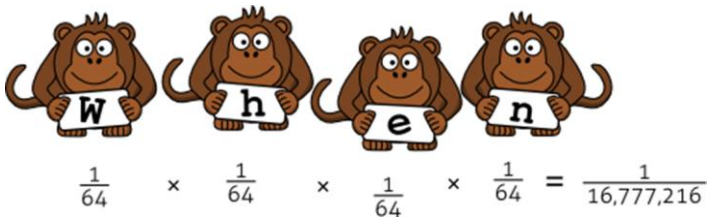
Genetic Algorithms.

Shakesperean monkey problem:



$$\frac{1}{64} \times \frac{1}{64} \times \frac{1}{64} \times \frac{1}{64} = \frac{1}{16{,}777{,}216}$$

Genetic Algorithms.



”To be or not to be that is the question”.

Guess 39 characters with brute force guesssing

- Take 1 million phrases per second and there

There is still not time enough in the known Universe -

(1/27)^39 is a very big number ...

Darwinian natural selection:

Use method inspired by Darwinian natural selection.

Principles:

- Heredity
- Variation
- Selection (Survival of the fittest).

Genetic Algorithms.

”To be or not to be that is the question”.

1. Create a population of N elements.

(Here, strings of characters 39 letters long)

2.

A. Calculate fitness.

(How many characters are right?).

B. Reproduction.

- Select parents based on fitness.

- Crossover, mutation.

Repeat with new population.

Genetic Algorithm,
Overview. Setup of our algorithm.

Genetic Algorithms.

1. Select parents for new pool based on fitness.

    We don't just take the best. We also need variation.

    Typically – 10 percent are just cloned. The rest are given propability of coming to new generation based on fitness.

2. Crossover ..

    We take material from two parents and combine to a new child

    ”To be hsjfhsdj ” + ”sdhfdsj or not”

    Gives ”To be or not” ,.,,,, ?

    There are many ways to do this – based on problem.

3. Mutation. Create variation.

Pool.

Pool :

”To be hsjfhsdj ”, ”sdhfdsj or not”, ””sdhfdsj
or dd dsfkd”, ”sdhfdsj or not that dfgdfh”

Osv. Osv.



Mutation.

1 % mutation gives some variation …

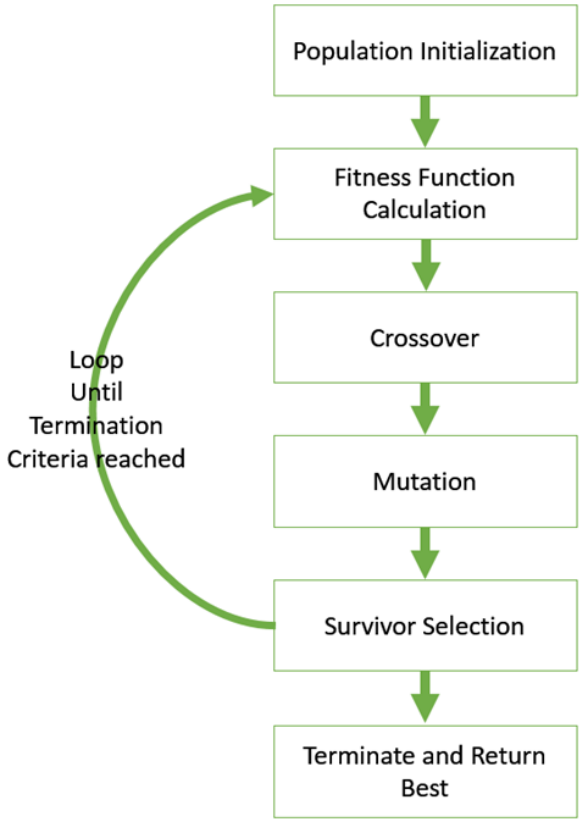
Two much mutation makes it impossible to settle on good solution.

No mutation – Can't find a solution.

Genetic Algorithms.

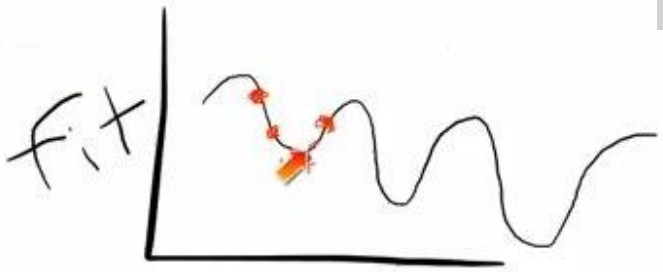
Bigger population is good, it gives us more to work with.

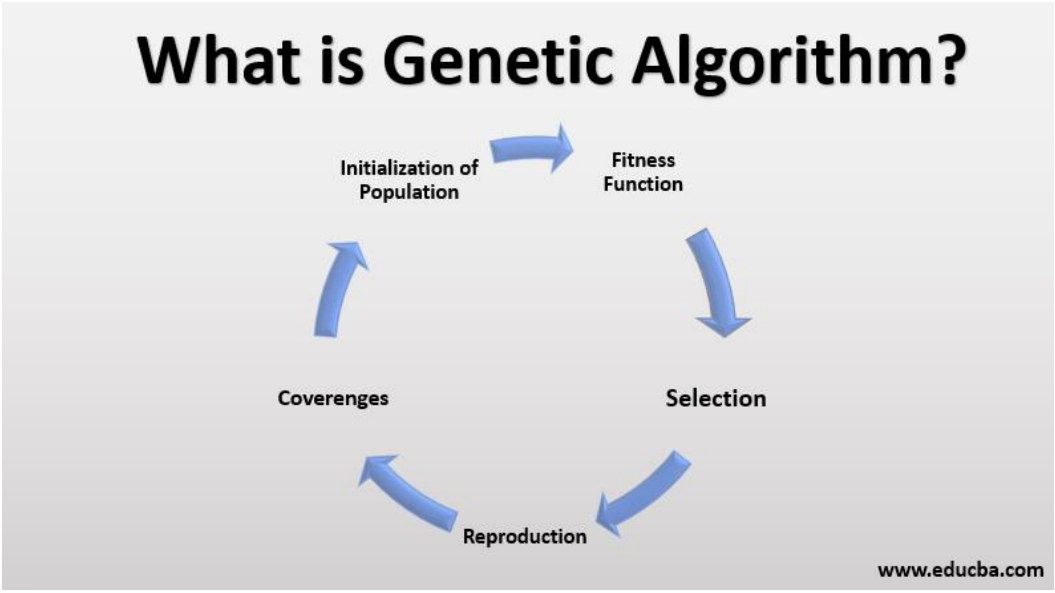
But is computational difficult to handle.



Genetic Algorithms.

Fitness – Searching for highest fitness.





Fitness Function

- A deterministic evaluation of a solution.
- Objective function to determine the merit of a solution.

0010	0101	1100	1000	1010	10
0110	0001	1101	0110	1111	14
0010	0111	1000	0011	1011	4
0000	1101	1101	0100	1110	1

Fitness Value

- More fitness -> Better solution -> More probability to survive

Evolutionære algoritmer.

Genetic Algorithms.

Our pool holds better and better solutions

”to be or not to be is the dsfghj”

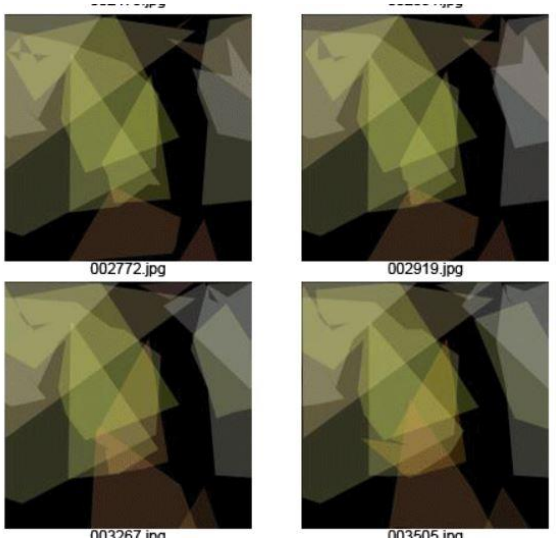
”to be or dsfhj is the ” ,,,

Etc.

And we eventually hit our target …..

Genetic Algorithms.

https://rogerjohansson.blog/2008/12/07/genetic-programming-evolution-of-mona-lisa/



Roger Johansson:

Mona Lisa:

https://rogerjohansson.blog/2008/12/07/genetic-programming-evolution-of-mona-lisa/

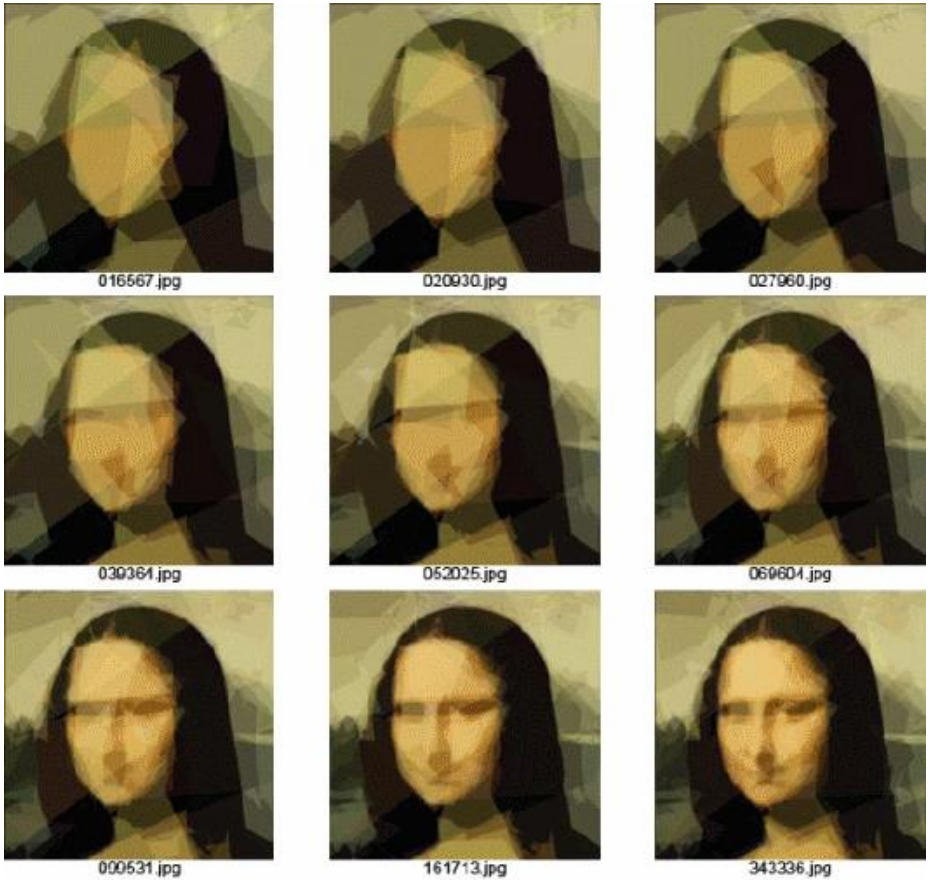
DNA now means:

Use the new DNA to render polygons onto a canvas

Fitness:

Compare the canvas to the source image.

Genetic Algorithms.

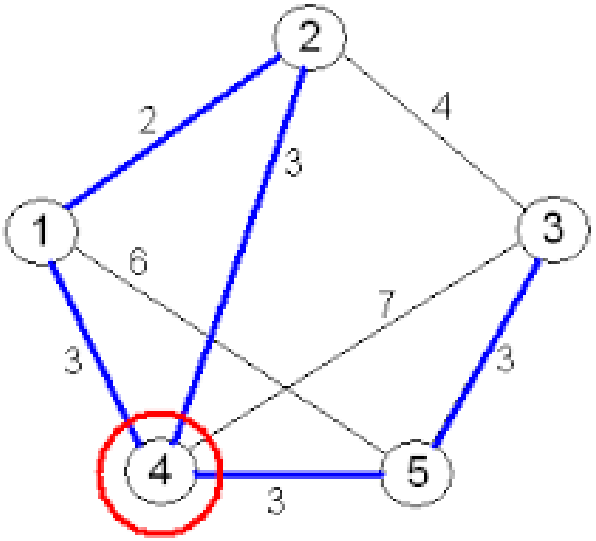


Evolutionære algoritmer.

ERHVERVSAKADEMI
AARHUS

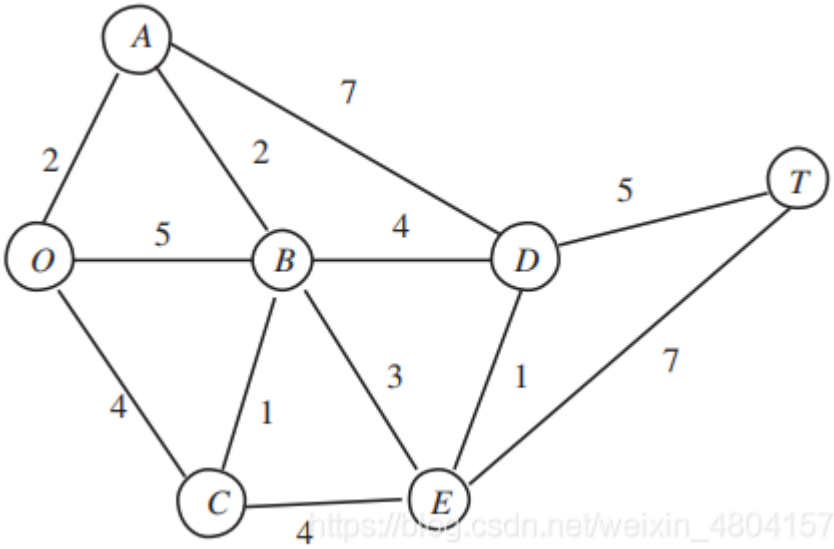
Travelling Salesman Problem

Very difficult – NP complete



Travelling Salesman Problem

Very difficult – NP complete



Diskuter i grupper:

Hvordan kunne man bruge GA til at løse TSP / Kunne man bruge GA til at løse TSP?

Travelling Salesman Problem

Again – we give a meaning to our DNA.

DNA:

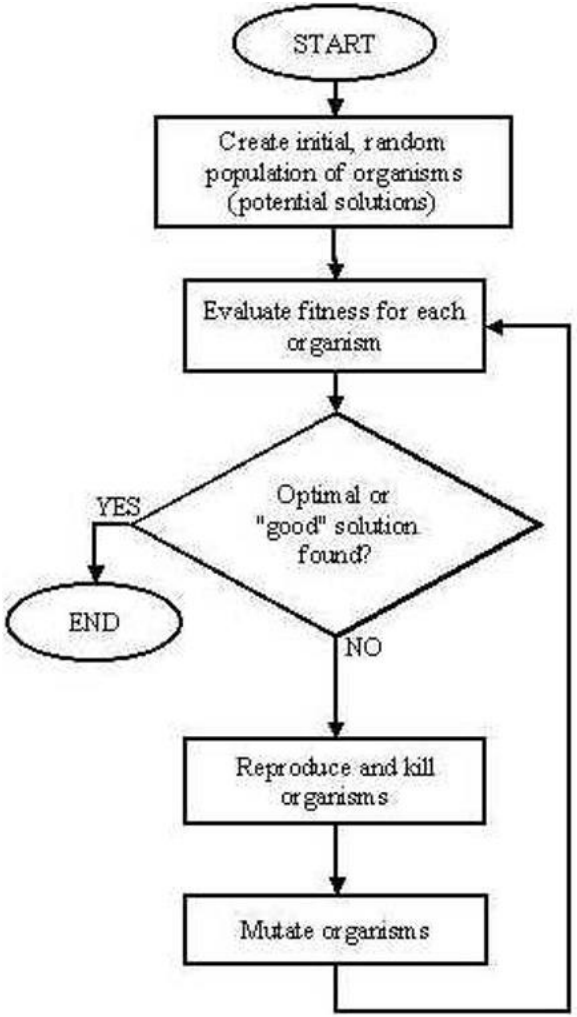
”jhsdfhsjzsdjdsjfh”

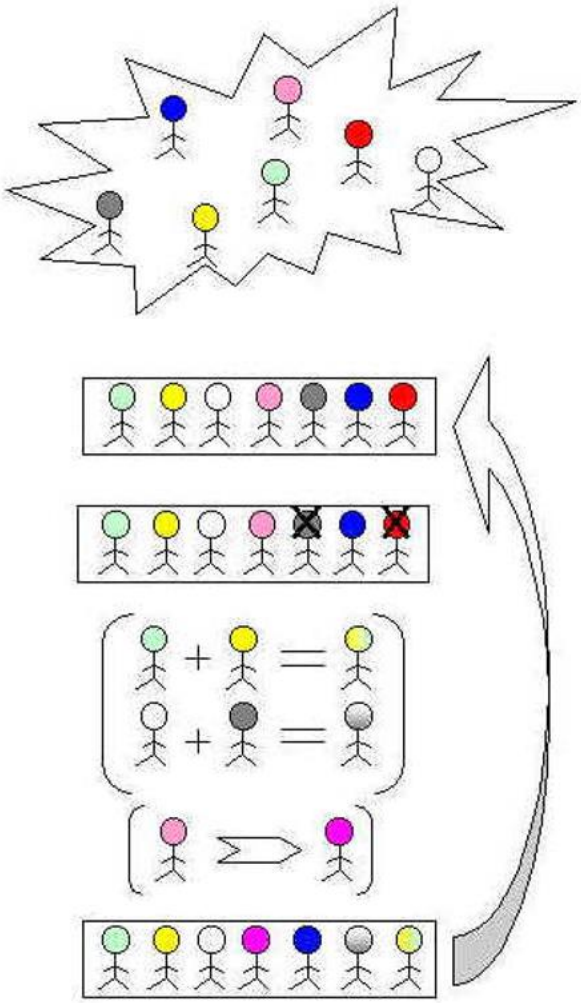
Tell us how we are going to visit the cities.

High fitness … we reach all cities, in shortest distance. …

Again – we make crossover, mutation … etc.

Genetic Algorithms.





Evolutionære algoritmer.

Genetic Algorithms.

Use of GA. Examples.

- Learning robot behavior using genetic algorithms.
- Image processing: Dense pixel matching.
- Learning fuzzy rule base using genetic algorithms.
- Code-breaking (search large solution spaces of ciphers for the one correct decryption).
- Finance and Economics
  Portfolio optimization.
- Airlines revenue management.
- Training artificial neural networks.
- Stochastic optimization.

Design of fitness function.

Tweaking

what if one fitness is

391.233

Another is 391.234

But the larger is really much better ???

We can use exponential functions – or something else that make the solutions stand out ...

Genetic Algorithms.

In words:

- Start with random population of organisms

- Evaluate fitness of organisms.

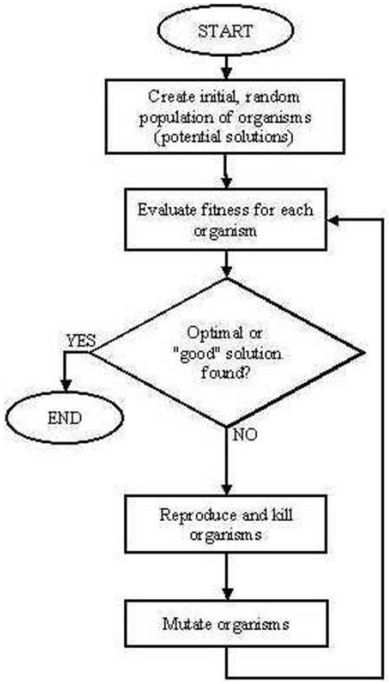
- Create a new population

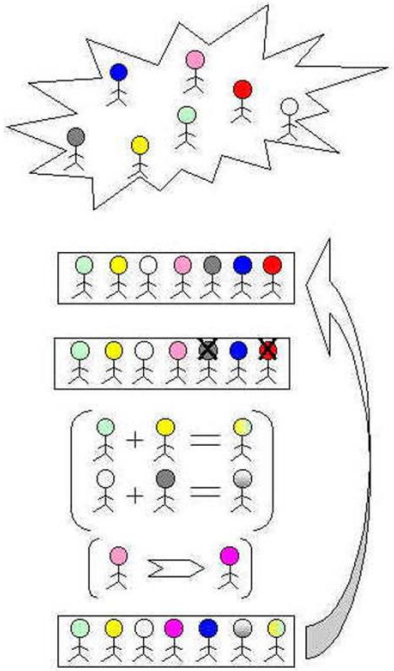
  Select parents based on fitness

  Create new children – crossover, mutations of DNA.

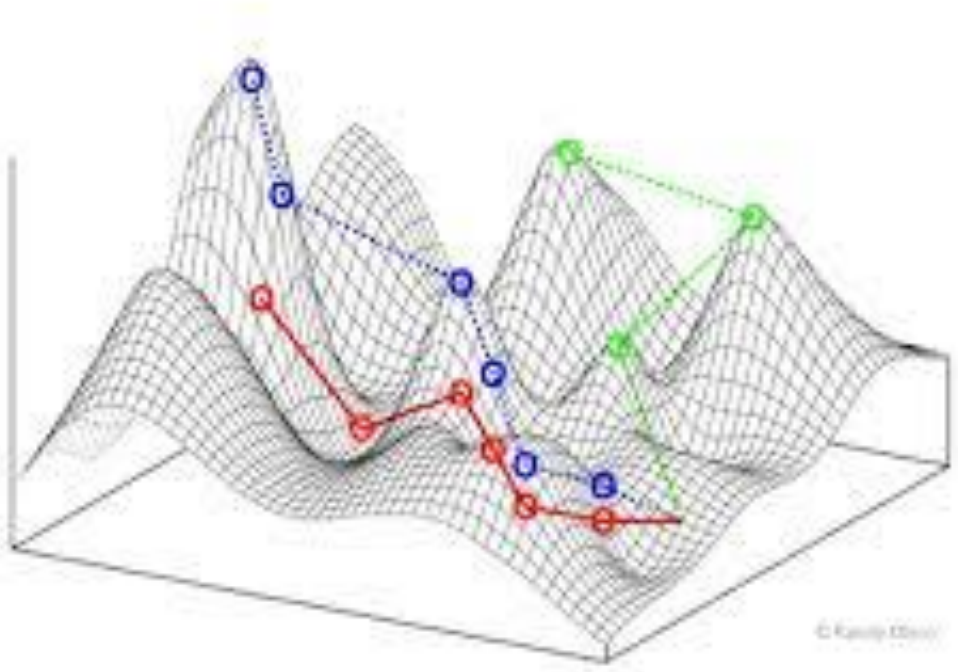
  Replace old population (perhaps letting the 10 percent best one stay on)

Repeat.





Fitness landscape



Resten af dagen – Opgaver.

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